

1. At 20-C, a brass cube has edge length 30cm. What is the increase in the surface area when it is heated from 20-C to 75-C?
2. An aluminium flagpole is 33 m high. By how much does its length increase as the temperature increases by 15 C-?
3. Find the change in volume of an aluminum sphere with an initial radius of 10 cm when the sphere is heated from 0.0-C to 100-C
4. At 20-C, a brass cube has edge length 30 cm. What is the increase in the surface area when it is heated from 20-C to 75-C?
5. Icebergs in the North Atlantic present hazards to shipping, causing the lengths of shipping routes to be increased by about 30% during the iceberg season. Attempts to destroy icebergs include planting explosives, bombing, torpedoing, shelling, ramming, and coating with black soot. Suppose that direct melting of the iceberg, by placing heat sources in the ice, is tried. How much energy as heat is required to melt 10% of an iceberg that has a mass of 200,000 metric tons? (Use 1 metric ton # 1000 kg.)
6. A pressure cooker contains water and steam in equilibrium at a pressure greater than atmospheric pressure. How does this greater pressure increase cooking speed?
7. The gauge pressure in your car tires is $2.50 \times 10^5 \text{ N/m}^2$ at a temperature of 35.0°C when you drive it onto a ferry boat to Alaska. What is their gauge pressure later, when their temperature has dropped to – 40.0°C?
8. Rubbing your hands together warms them by converting work into thermal energy. If a woman rubs her hands back and forth for a total of 20 rubs, at a distance of 7.50 cm per rub, and with an average frictional force of 40.0 N, what is the temperature increase? The mass of tissues warmed is only 0.100 kg, mostly in the palms and fingers.
9. A 0.250-kg block of a pure material is heated from 20.0°C to 65.0°C by the addition of 4.35 kJ of energy. Calculate its specific

heat and identify the substance of which it is most likely composed.

10. Following vigorous exercise, the body temperature of an 80.0-kg person is 40.0°C . At what rate in watts must the person transfer thermal energy to reduce the body temperature to 37.0°C in 30min, assuming the body continues to produce energy at the rate of 150W? 1 watt = 1 joule/second or $1\text{ W} = 1\text{ J/s}$

NOTE use the various constants from the lecture materials and other related textbooks of your choice